

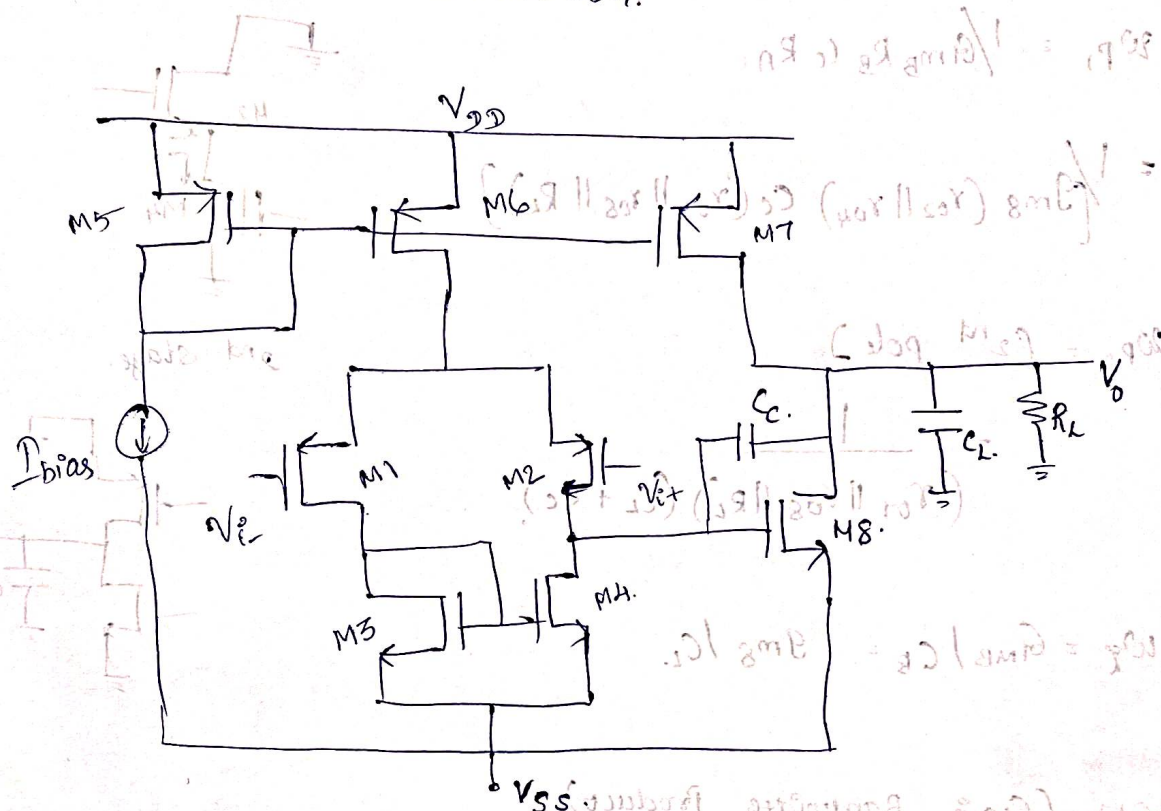
PRELAB: Two stage Amplifier Design

With 3rd Order Driven Statistical
Corner Extraction.

ECEN 607: Advanced Analog Circuit Technology

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Given Design Specifications:-

$$V_{DD} = 0.9V$$

$$V_{SS} = -0.9V$$

$$A_{v0} > 40dB$$

$$GBW > 2MHz$$

$$PM > 45^\circ$$

$$\text{Output swing} > \pm 0.5V$$

$$C_L = 30pF$$

$$R_L = 10k\Omega$$

1. Expressions for DC Gain (A_{V_0}), Dominant pole (p_1), Second pole (p_2), Zero (z), Gain bandwidth product (GBW), Phase Margin (PM)

$$A_{V_0} = G_{mR_{out}} = g_{m1,2} (r_{o2} \parallel r_{o4}) \cdot g_{m8} (r_{o1} \parallel r_{o8} \parallel R_L)$$

Dominant pole: p_1

$$\omega_{p1} = 1 / G_{mB} R_B C_C R_A$$

$$= 1 / [g_{m8} (r_{o2} \parallel r_{o4}) C_C (r_{o1} \parallel r_{o8} \parallel R_L)]$$

ω_{p2} (2nd pole)

$$= \frac{1}{(r_{o1} \parallel r_{o8} \parallel R_L) (C_L + C_C)}$$

$$\omega_z = G_{mB} / C_B = g_{m8} / C_L$$

GBW (Gain Bandwidth Product)

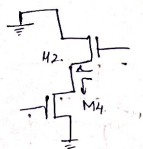
$$= W_{GBW} / 2\pi$$

$$= G_{mA} / 2\pi$$

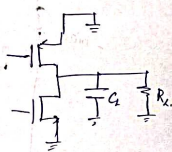
$$= g_{m1,2} / 2\pi C_C$$

$$\text{Phase margin: } PM = 90^\circ - \tan^{-1}(\omega_{GBW} / \omega_{p2}) - \tan^{-1}(\omega_{GBW} / \omega_z)$$

Doing half circuit analysis for the 1st stage.



2nd stage.



$$A_{V_0} = \frac{1}{\lambda_n V_{eff1,2} \lambda_p V_{eff8}} > 40 \text{ dB}$$

Finding: Choosing Miller Capacitance to be, $C_C = 10 \text{ pF}$

Finding required g_{m1} from GBW ,

$$GBW = \frac{g_{m1}}{2\pi C_C}, \text{ Let } GBW = 2.5 \text{ MHz}$$

$$g_{m1} = 157 \times 10^{-6} \text{ A/V}$$

Tail current. Start,

Let $V_{eff1,2} = 1 \text{ V}$.

$$g_{m1,2} = \frac{2I_D}{V_{eff}}$$

$$g_{m1,2} = \frac{I_{tail}}{1 \text{ V}}$$

$$I_{tail} = 157 \mu\text{A}$$

$$I_D = 78.5 \mu\text{A}$$

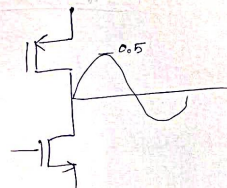
From the output swing, calculating the over-drive, V_{eff} voltage of the transistors:

$$\text{Let } (V_{eff})_g = 0.35$$

and

$$\text{Let } (V_{eff})_n < 0.9 - 0.55$$

$$(V_{eff})_n = 0.25$$



$$(V_{GS})_8 - (V_{th})_8 = 0.55$$

$$(V_{GS})_8 = 0.55 + 0.46$$

$$(V_{GS})_8 = 1.01$$

$$(V_{GS})_8 - (V_S)_8 = 1.01$$

$$V_G = 1.01 + V_S$$

$$= 1.01 - 0.9$$

$$(V_G)_8 = 0.11$$

$$(V_{ov})_4 = V_{GS} - V_{th}$$

$$(0.11 + 0.9) - 0.46$$

$$(V_{ov})_4 = 0.55$$

$$(V_{ov})_6 = V_{GS} - V_t$$

$$V_{eff1} = 0.25$$

$$V_{GS7} - V_t = 0.25$$

$$V_{GS7} = 0.25 + 0.43$$

$$V_{GS7} = 0.68$$

$$V_{G7} - V_{S7} = 0.68$$

$$V_{G7} = 0.68 + V_{S7}$$

$$= 0.68 + 0.9$$

$$V_{G7} = 1.58$$

$$(V_{ov})_8 = V_{GS} - V_t$$

$$= (1.58 - 0.9) - 0.43$$

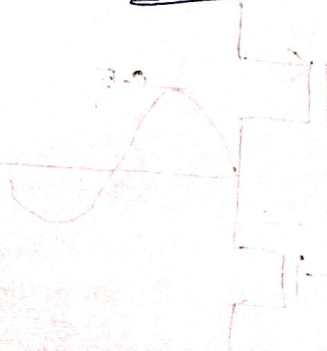
$$(V_{ov})_6 = 0.25$$

$$1.8 - V_{eff6} - V_{eff1} - V_{eff4} = 0$$

$$1.8 - 0.25 - V_{eff1} - 0.55 = 0$$

$$V_{eff1} \leq 1V$$

$$V_{eff} = 0.8V$$



$$V_{eff,1,2} = 0.8 V \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} 78.5 \mu A$$

$$V_{eff,3,4} = 0.55$$

$$V_{eff,8} = 0.55$$

$$V_{eff,6} = 0.25$$

$$V_{eff,7} = 0.25$$

$$V_{eff,5} = 0.25$$

$$151 \mu A$$

For, $V_{eff,1,2}$

$$I_D = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{eff})^2$$

$$78.5 \times 10^{-6} = \frac{47.2 \times 10^{-6}}{2} \left(\frac{W}{L} \right) (0.8)^2$$

$$\frac{78.5}{15.104} = \frac{W}{L}$$

$$5.197 = \frac{W}{L}$$

$$\boxed{\frac{2 \times 0.5}{0.18} = \left(\frac{W}{L} \right)_{1,2}}$$

$$(I_D)_{3,4} = 78.5 \mu A$$

$$78.5 \mu A = \frac{W}{L} (0.55)^2 \times \frac{47.2}{2} \times 10^{-6}$$

$$\frac{78.5}{7.139} = \frac{W}{L}$$

$$4 \times \left(\frac{0.5}{0.18} \right) = \frac{W}{L}$$

$$I_{D,8}$$

$$151 \mu A = \frac{167}{2} \times \left(\frac{W}{L} \right) (0.55)^2$$

$$151 = 25.25 \frac{W}{L}$$

$$\frac{151}{25.25} = \frac{W}{L}$$

$$\left(2 \times \frac{0.5}{0.18} \right) = \left(\frac{W}{L} \right)_8$$

$$D_{59.7} = \frac{41.2 \times 10^{-6}}{2} \left(\frac{\omega}{L} \right) (0.25)^2$$

$$157 \mu A = \frac{41.2}{2} \times (0.25)^2 \frac{\omega}{L}$$

$$157 \mu A = 1.415 \frac{\omega}{L}$$

$$106.44 = \frac{\omega}{L}$$

$$38 \times \frac{0.5}{0.18} = \frac{\omega}{L}$$

finding phase Margin and compensation

To find A_{v0} ,

$$g_{m1,2} = 157 \mu S$$

$$g_{m8} = \frac{D_D}{V_{eff}}$$

$$= \frac{157 \mu A}{0.55}$$

$$= 285.45 \mu S$$

$$r_{01} = \frac{D_D}{V_{eff}}$$

$$r_{02} = \frac{78.5 \mu A}{0.55}$$

$$r_{02} = r_{ds} = \frac{1}{\lambda_p D_D}$$

$$r_{02} = \frac{1}{\lambda_p D_{1,2}}$$

$$= \frac{1}{78.5 \mu A \times 289.27 \times 10^{-3}}$$

$$r_{02} = 44037.935$$

$$r_{04} = \frac{1}{263 \times 10^{-3} \times 78.5 \mu A}$$

$$r_{04} = 48436.705$$

$$r_{01} = \frac{1}{\lambda_p D_D}$$

$$r_{01} = \frac{1}{10^{-6} (157) \times 289.27 \times 10^{-3}}$$

$$r_{01} = 22018.967$$

$$r_{08} = \frac{1}{\lambda_n D_D}$$

$$= \frac{1}{10^{-6} (263) (157) \times 10^{-3}}$$

$$r_{08} = 24218.35267$$

$$A_{v0} = \frac{1}{0.263 \times 0.28927 \times 0.55 \times 0.55} > 40dB$$

$$A_{v0} \approx 44$$

$$\frac{1}{0.263 \times 0.28927 \times 0.55 \times 0.55} > 450$$

$$\frac{1}{0.263 \times 0.28927 \times 0.55 \times 150} > x$$

$$\frac{1}{\lambda_{1,2}} \leq 0.159$$

$$\text{Adjusting } V_{eff1,2} = 0.159$$

$$D_D = \frac{I_{nCor}}{2} \frac{\omega}{L} (V_{eff})^2$$

$$78.5 \times 10^{-6} = \frac{41.2 \times 10^{-6}}{2} \times \frac{\omega}{L} (0.159)^2$$

$$88 \times \left(\frac{\omega}{L} \right) = \frac{\omega}{L}$$

$$\omega_{p1} = \frac{1}{\left[g_{m8} (r_{02} \parallel r_{04}) C_c (r_{01} \parallel r_{08} \parallel R_L) \right]}$$

$$= \frac{1}{\left(\frac{1}{285.45} \right) (44037.935 \parallel 48436.705) \times 10 \times 10^{-12} \times (22018.967 \parallel 24218.35267 \parallel 10 \times 10^3)}$$

$$\frac{1}{285.45 \times 10^{-6} \times 23066.35 \times 10 \times 10^{-12}} = 5356$$

$$= \frac{1}{\dots}$$

$$\omega_{p1} = 2.83 \text{ MHz}$$

$$\omega_{p2} = \frac{1}{r_{o1} \parallel r_{o2} \parallel R_L (C_L + C_c)}$$

$$= \frac{1}{5356.003 (10 \text{ pF} + 30 \text{ pF})}$$

$$= 4.66 \text{ MHz}$$

$$\boxed{\omega_{p2} = 4.66 \text{ MHz}}$$

$$GBW = (15.7 \times 10^6) \times 2\pi \times 80 \times 10^{-12}$$

$$= 2.5 \text{ MHz}$$

$$PM = 90^\circ - \tan^{-1}(2.5/4.6) - \tan^{-1}(2.5/9.5)$$

$$\boxed{PM = 46.733^\circ}$$

$$\Rightarrow > 45^\circ \text{ (satisfied)}$$

$$\boxed{GBW = 2.5 \text{ MHz}}$$

$$\omega_z = g_{m8} / C_L$$

$$= 285.45 \times 10^{-6} / 30 \times 10^{-12}$$

$$= 9.5 \text{ MHz}$$

$$\boxed{PM = 30^\circ}$$